Corrie: Design of a Social Campus Guide Robot

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Abstract

This paper presents the design of a campus guide robot created to assist students, professors, staff, and visitors with navigation and information requests on a university campus. The robot is implemented on the Misty II platform and combines two coordinated pipelines for navigation and communication.

The navigation pipeline uses simultaneous localization and mapping to autonomously plan safe indoor routes, while the communication pipeline supports conversational interaction that includes remarks about campus landmarks and building history. During guidance, the robot engages users through dialogue, pointing out features such as reception desks, vending machines, or posters and sharing context about the university environment.

Through interaction scenarios that compare human guides and robot guides, the design illustrates how social supports, such as conversation, landmark-based guidance, and expressive feedback, can transform a functional navigation system into a human-centered companion that complements existing human help desks.

Keywords

Human-Robot Interaction, Campus Guide Robots, Design, User Experience, Interaction Scenarios

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1 Introduction

Navigating a large university campus can be confusing, especially for new students, visiting parents, or guests attending meetings or events. Although maps and human help desks are available, they face limitations in accessibility and availability. Help desks are typically few in number and concentrated in specific locations, making them difficult to access for users who are already lost or in different buildings. Additionally, human help desk staff must remain at their stations to assist multiple visitors, limiting their ability to accompany users to destinations.

A campus guide robot can address these limitations by complementing human help desks. Our design focuses on creating a reliable and approachable guide that mimics the behaviors of effective human guides. By replicating how human guides communicate naturally, point out landmarks, and share campus information, the robot creates interactions that feel natural and supportive rather than purely functional.

The overall goal is to design a campus guide robot that extends the reach of human help desks by providing accessible, on-demand navigation support across campus buildings. By drawing on social behaviors common to effective human guides, including conversation, landmark-based guidance, and contextual commentary, the robot offers assistance that feels welcoming and supportive rather than purely technical. In doing so, it helps help desks offload routine wayfinding requests, maintain availability for complex questions, and provide a more consistent level of support to users regardless of where they are on campus.

2 Design Goals and Principles

The design of the campus guide robot is organized around four core goals: satisfaction, inclusion, trust, and independence. These goals shape both the technical pipelines and the interaction scenarios so that the robot not only completes navigation tasks but also contributes to a positive user experience.

Satisfaction. The robot aims to provide smooth, understandable, and pleasant interactions that leave users feeling supported and confident about where they are going. Clear explanations of routes, timely progress updates, and expressive feedback during guidance all contribute to satisfaction.

Inclusion. The system is intended to support a wide range of campus visitors, including students, staff, faculty, parents, and guests who may be unfamiliar with the environment. By offering approachable conversation, landmark-based explanations, and campus context, the robot treats each user as a welcome participant in campus life.

Trust. Predictable behavior and consistent communication help users feel comfortable following the robot or relying on its verbal directions. The robot explains what it is doing, describes upcoming turns and landmarks, and maintains steady, legible motion so that users can easily anticipate its actions.

Independence. Users can choose whether they want to walk with the robot or simply receive directions and continue on their own. Landmark-based guidance and contextual commentary are designed to give users enough information to navigate independently if they prefer, while still supporting escorted guidance when desired.

Together, these goals guide decisions about how the robot moves, when it communicates, what information it shares, and how it is integrated with existing help desk services.

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3 Robot Design

The campus guide robot is implemented on the Misty II platform, which provides a mobile base, cameras, time-of-flight sensors, microphones, and expressive LED eyes. The design integrates navigation and communication capabilities so that the robot can move safely through indoor spaces while maintaining engaging, human-like interaction.

Misty's sensing and mobility support the navigation pipeline by enabling SLAM-based mapping, obstacle detection, and route execution in hallways, lobbies, and other indoor environments. Its expressive head, eye animations, and audio output support the communication pipeline, allowing the robot to greet users, comment on landmarks, and provide progress updates in a personable manner. By coordinating motion with gaze, speech, and simple gestures such as turning toward a reception desk, the robot presents itself as a social guide rather than a purely technical tool.

The system comprises two main design pipelines that work together to create a seamless user experience.

1. Navigation Pipeline: This component autonomously maps indoor spaces using simultaneous localization and mapping (SLAM), identifies fixed obstacles, and plans efficient routes to destinations such as classrooms, cafeterias, or offices. The robot tracks its own position, observes its surroundings through cameras and time-of-flight sensors, and adjusts its movement whenever obstacles are detected. The system operates autonomously as the primary mode, with a Wizard-of-Oz (WoZ) setup available only when required for testing or exceptional circumstances.

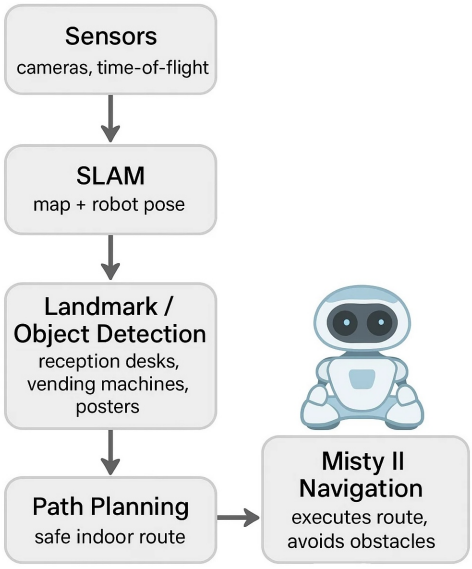


Figure 1: Navigation Pipeline

2. Communication Pipeline: This component manages all conversational behavior. The robot engages users during navigation by sharing information about landmarks, building history, and campus features. It detects the user's language and direction of speech, maintains visual attention, and processes input through a retrieval-based model that provides contextually appropriate responses. To maintain clarity, the robot focuses on one user at a time.

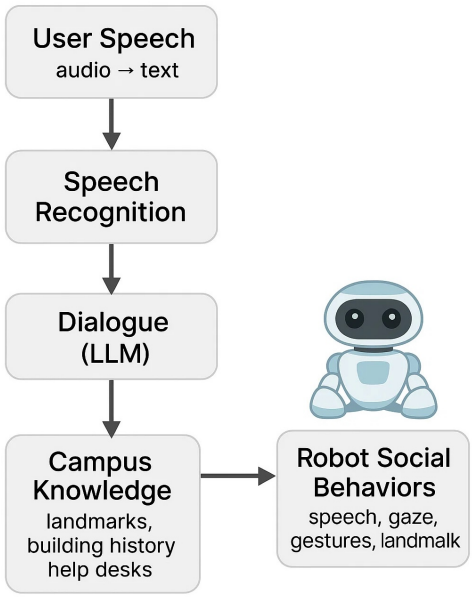


Figure 2: Communication Pipeline

4 Interaction Scenarios

The design supports multiple on-campus interaction scenarios that demonstrate how the robot complements human help desks. To understand why we designed the robot to mimic human guide behavior, we first examine how human guides provide navigation assistance, then show how the robot replicates these behaviors to deliver similar social supports and user experiences.

4.1 Scenario: Human Guide Behavior

Consider how a human guide would help someone navigate on campus. A visitor approaches a help desk asking for directions to a specific room. The human staff member, unable to leave their post due to other potential visitors, might describe the route verbally. However, if the help desk is staffed by multiple people or has fewer visitors at the moment, a human guide could accompany the visitor.

As they walk, the human guide engages in light conversation, pointing out interesting features. For example, when passing an empty reception desk, the guide might slow down, orient toward the desk, point at it, and say in an upbeat, informative tone, "Here, from Monday through Friday, Enrique sits from 7am to 6pm. You can ask him about anything!" The guide shares context about who works where, mentions interesting posters, points out vending machines, or shares building history, making the journey informative and engaging rather than just functional. This relaxed, welcoming approach helps visitors feel included and welcome, builds trust through consistent helpful behavior, and creates a satisfying experience that goes beyond simple wayfinding.

4.2 Scenario: Robot Guide Behavior - Mimicking Human Guides

The robot guide is designed to replicate these human guide behaviors because they create positive user experiences that contribute to satisfaction, inclusion, trust, and independence. When a visitor approaches a human help desk, the staff member can dispatch a robot guide for navigation tasks, allowing the human to remain available for other visitors. The robot then accompanies the user, keeping up a conversational, interactive tone similar to how a human guide would speak.

As the robot moves along the route, it identifies landmarks using either locations pinned on the SLAM map or simple object detection over camera data. Map pins make key wayfinding anchors, such as reception desks or vending machines, reliably available, while perception lets the robot comment on more dynamic elements like posters, giving a lightweight hybrid approach without adding extra complexity to the navigation system. When reaching a notable feature like a reception desk, the robot mimics the human guide's behavior: it slows to a halt, orients itself toward the landmark, and points at it using an arm gesture. In a warm tone and with expressions matching the human guide's approach, it says, "Here, from Monday through Friday, Enrique sits from 7am to 6pm. You can ask him about anything!" By replicating this interaction pattern, the robot creates the same inclusive, welcoming experience that helps users feel part of the campus community. The robot adds brief remarks or shares interesting building history, creating an engaging experience that mirrors what users would expect from a human guide.

The robot provides step-by-step spoken directions using landmarks, following the same approach human guides use. For example: "We'll head down this hallway and take a left at the vending machines up ahead. Those machines actually have some of the best snacks on campus! Once we turn left, we'll see the library entrance on the right - that's where you'll find quiet study spaces if you need them. Our destination is just past the library, the third door on the left." This landmark-based guidance, combined with conversational commentary, gives users enough information to navigate independently if they choose, supporting their autonomy while also offering the option of being accompanied, the same flexibility human guides provide.

During the journey, the robot periodically communicates progress in an encouraging way: "We're about halfway there! Just a bit further and we'll see the CS building's main lobby." This mirrors how human guides provide reassurance and keep visitors informed. The friendly tone and landmark-based guidance make the interaction feel natural and supportive, building trust that the robot knows where it is going and contributing to overall satisfaction with the navigation experience, just as consistent communication from a human guide would.



Figure 3: Human-Robot Interaction

4.3 Scenario: Routine Tasks and Help Desk Efficiency

This scenario illustrates how mimicking human guide behavior enables robots to complement human help desks effectively. A visitor approaches a help desk in the library seeking directions to a computer lab. The help desk staff member recognizes this as a routine navigation request that can be handled by the robot. Instead of leaving their post or providing verbal directions that might be confusing, the staff member dispatches a robot guide.

The robot greets the visitor in an easygoing manner and asks, "Would you like me to walk with you there? I can point out some interesting spots along the way!" This mirrors how a human guide would offer assistance, giving users the choice to accept or decline and supporting their independence in deciding how they want help. If the visitor agrees, the robot begins navigation while engaging in small talk, just as a human guide would. It might comment on a poster they pass: "Oh, check out this poster about the robotics club - they meet every Thursday if you're interested in that kind of thing!" This informal interaction makes the experience pleasant and engaging, creating the same satisfying, inclusive experience users would have with a human guide.

This scenario demonstrates how robots complement human help desks by handling routine navigation tasks, allowing human staff to remain available for complex questions, troubleshooting, or situations requiring human judgment and empathy. By replicating human guide behaviors, such as light conversation, landmark pointing, and friendly commentary, the robot provides the same social support users expect, building trust and satisfaction while maintaining the inviting, helpful experience that makes campus navigation feel welcoming rather than mechanical.

4.4 Indoor Environment Limitations

All scenarios take place indoors, such as in libraries, computer science buildings, labs, or corridors. This controlled environment is suitable given Misty's capabilities and ensures safety, smooth movement, and realistic user interactions. Indoor spaces provide consistent landmarks like reception desks, vending machines, posters, and room signage that the robot can reliably identify and discuss.

5 Complementing Human Help Desks

A central focus of this design is how the campus guide robot complements rather than replaces human help desks. This complementary relationship addresses key limitations of traditional help desk services while enhancing their effectiveness.

5.1 Accessibility and Distribution

Human help desks are limited in number and spread across campus, making them difficult to access for users who are already lost or located in different buildings. By deploying robots that can be stationed throughout campus buildings, the design significantly improves accessibility. Robots can be placed in multiple locations, such as libraries, CS buildings, lab areas, and main lobbies, ensuring that navigation assistance is available wherever users need it, not just at centralized help desk locations.

5.2 Dispatching Robots from Help Desks

When visitors approach a human help desk with routine navigation requests, staff members face a dilemma: leaving their post to guide someone means they are unavailable for other visitors who may arrive. Our design addresses this by allowing help desk staff to dispatch robot guides for wayfinding tasks.

A visitor arrives at a help desk asking for directions to a classroom. The staff member, recognizing this as a routine task that does not require human judgment, activates a nearby robot guide. The robot approaches, greets the visitor in a welcoming tone, and offers to accompany them: "Hi! I can walk with you to that classroom and show you some cool spots along the way." The staff member remains at their post, available to assist other visitors or address complex questions that require human assistance.

This dispatching capability makes human help desks more efficient. Staff members can focus on tasks that truly benefit from human interaction, such as answering complex questions, troubleshooting issues, providing personalized recommendations, or handling situations that require empathy and judgment. Meanwhile, robots handle routine navigation tasks, expanding the help desk's reach and availability. Users still receive the friendly, informative guidance they need, building trust in both the robot and the help desk system as a whole. The robot's consistent, approachable communication style ensures all users feel included and comfortable, regardless of whether they're interacting with a robot dispatched from a help desk or a human staff member.

5.3 Social Supports Through Mimicking Human Behavior

The robot's design emphasizes providing social supports by mimicking the behaviors that make human guides effective. Through casual conversation, lighthearted comments, and sharing information about campus landmarks and history, the robot creates an engaging, supportive experience that mirrors what users receive from human guides. This social engagement builds trust and makes users more comfortable seeking assistance, whether directly from a robot or when one is dispatched by a help desk.

By replicating human guide behaviors such as an easygoing tone, landmark pointing, progress updates, and friendly commentary, the robot delivers the same social supports that contribute to user satisfaction, inclusion, trust, and independence. The combination of improved accessibility (robots available throughout campus) and help desk efficiency (dispatch capability) creates a complementary system where robots extend the reach and effectiveness of human help desks while maintaining the personal, supportive experience users expect from campus assistance services. Users receive the same quality of interaction whether they're guided by a human or a robot, ensuring consistent experiences that build trust in the overall help desk system.

6 Conclusion

This design paper presents Corrie, a social campus guide robot that combines autonomous indoor navigation with expressive, landmark-based interaction. By coordinating navigation and communication pipelines on the Misty II platform, the robot can safely lead users through campus buildings while providing conversational explanations of landmarks and campus context. The interaction scenarios and help desk integration illustrate how robot guides can handle routine wayfinding tasks, extend the reach of sparse help desks, and maintain an approachable, human-centered experience for visitors, students, and staff across the campus.

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